DANIEL LEVINE

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Yale School of Medicine
Department of Internal Medicine
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Experience

Yale-Boehringer Ingelheim Biomedical Data Science Postdoctoral Fellow — June 2025 - Present
van Dijk Lab at Yale University

Postdoctoral Associate — May 2023 - May 2025
van Dijk Lab at Yale University

Machine Learning Engineer — January 2021 - May 2022
Avanan/Check Point Software Technologies

Education

Pennsylvania State University — May 2020
Ph.D. in Mathematics
Advised by Jack Huizenga

Stony Brook University — May 2014
B.S. in Mathematics with Honors

Awards and Honors

Yale-Boehringer Ingelheim Biomedical Data Science Postdoctoral Fellowship — 2025
Stellar Abstract Award, Program in Quantitative Genomics Conference — 2024
Jack and Eleanor Pettit Scholarship in Science, Penn State University — 2016-2017

Publications and Preprints

* denotes co-first author

(1) **D. Levine**, D. van Dijk, K. Kay *Self-Modifying Neural Networks that Zero-Shot Infer and Continually Learn* arXiv 2026, https://arxiv.org/abs/2610.40757v1

(2) S.A. Rizvi, **D. Levine***, A. Patel, S. Zhang, E. Wang, C. Perry, I. Vrkic, N.M. Constante, Z. Fu, S. He, D. Zhang, C. Tang, Z. Lyu, R. Darji, C. Li, E. Sun, D. Jeong, L. Zhao, J. Kwan, D. Braun, B. Hafler, H. Chung, R.M. Dhodapkar, P. Jaeger, J. Ishizuka, S. Azizi, D. van Dijk, *Scaling Large Language Models for Next-Generation Single-Cell Analysis* (Appearing in Nature) 2026, https://www.biorxiv.org/content/10.1101/2025.04.14.648850v4

(3) C. Perry, A. Frey, Y. Fei, J. Wang, P. Costa, H.N. Amin, S. Ghadermarzi, **D. Levine**, W. Tong, A. Koda, M. Mackie, M. He, Y. Nie, K. Clulo, F. Ouerghi, J. Wei, T. Cordero Dumit, M. Yaskolko, M. Ding, A. Caldera, O. Kyrysyuk, L. Lum, J.W. Allen, W. Guo, A. Naqash, A. Elliot, A. Vanderwalde, M. Capelletti, T. Adeyelu, S.T. Barry, M. Hugaboom, A. Bacchiocchi, H. Kluger, M. Bosenberg, A. Iwasaki, D. Braun, J. Clune, D. van Dijk, K. Olino, J.J. Ishizuka *Perturbational single-cell profiling of patient tumors defines lineage- and context-specific programs of innate immune resistance* bioRxiv 2025 https://www.biorxiv.org/content/10.64898/2025.12.09.693245v1

(4) Y. Zhang, S. He, **D. Levine**, L. Zhao, D. Zhang, S.A. Rizvi, E. Zappala, R. Ying, D. van Dijk, *Non-Markovian Discrete Diffusion with Causal Language Models* NeurIPS 2025, https://neurips.cc/virtual/2025/loc/san-diego/poster/115857

(5) S. He, **D. Levine***, I. Vrkic, M. Bressana, D. Zhang, S. Rizvi, Y. Zhang, E. Zappala, D. van Dijk, *CaLMFlow: Volterra Flow Matching using Causal Language Models*, arXiv 2024, https://arxiv.org/abs/2410.05292

(6) **D. Levine**, S.A. Rizvi, S. Lévy, N. Pallikkavaliyaveetil, D. Zhang, X. Chen, S. Ghadermarzi, R. Wu, Z. Zheng, I. Vrkic, A. Zhong, D. Raskin, I. Han, A.H.O. Fonseca, J.O. Caro, A. Karbasi, R.M. Dhodapkar, and D. van Dijk, *Cell2sentence: Teaching large language models the language of biology*, ICML 2024, https://icml.cc/virtual/2024/poster/34580

(7) J.O. Caro, A.H.O. Fonseca, C. Averill, S.A. Rizvi, M. Rosati, J.L. Cross, P. Mittal, E. Zappala, **D. Levine**†, R.M. Dhodapkar, I. Han, A. Karbasi, C.G. Abdallah, and D. van Dijk, *BrainLM: A foundation model for brain activity recordings*, ICLR 2024 †*on preprint*, https://www.biorxiv.org/content/10.1101/2023.09.12.557460v2

(8) E. Zappala, **D. Levine**, S. He, S. Rizvi, S. Lévy, D. van Dijk, *Operator Learning Meets Numerical Analysis: Improving Neural Networks through Iterative Methods* arXiv 2023, arxiv.org/abs/2310.01618

(9) **D. Levine** and S. Zhang, *Brill-Noether and existence of semistable sheaves on del Pezzo surfaces*, Annales-Institut Fourier (2022), https://aif.centre-mersenne.org/articles/10.5802/aif.3619/

Invited Talks

(Upcoming) Joint Mathematics Meeting	February 2027
Yale-Boehringer Ingelheim Colloquium	September 2026
Yale AI Symposium	April 2026
Yale-Boehringer Ingelheim Day	April 2026
Prescient Design, Genentech	November 2025
Yale-Boehringer Ingelheim Colloquium	September 2025
Rutgers Math LLM Seminar	December 2024
AstraZeneca, Centre for AI	November 2024
IBM Research Accelerated Discovery Group Seminar	October 2023
Yale Single Cell Symposium	October 2023
University of Arkansas Algebra Seminar	April 2019
FRG Workshop on Moduli Spaces of Sheaves and Bridgeland Stability UIC	December 2018
Penn State Algebra and Number Theory Seminar	December 2018
Indiana University Algebra Seminar	November 2018

Posters

Self-directed Learning in Plastic Feedforward Networks that Zero-Shot Generalize and Continually Learn
Yale AI Symposium — April 2026
Yale University

Non-Markovian Discrete Diffusion with Causal Language Models
NeurIPS — December 2025
San Diego

Cell2Sentence: Teaching Large Language Models the Language of Biology
ICML — July 2024
Vienna

FDS Workshop: Theory and Practice of Foundation Models — October 2023
Yale University

Existence of a semi-stable sheaf on a del Pezzo surface of degree ≥ 5
Arithmetic & Algebraic Geometry in Ann Arbor — August 2019
University of Michigan
Algebraic Geometry Northeastern Series (AGNES) — March 2019
University of Massachusetts Amherst
Midwest Algebraic Geometry Graduate Conference (MAGGC) — March 2019
University of Illinois at Chicago
Brill-Noether for del Pezzos
Midwest Algebraic Geometry Graduate Conference (MAGGC) — May 2018
University of Illinois at Chicago

Conferences/Workshops Attended

(Upcoming) Joint Mathematics Meetings — January 2027
Chicago
Cognitive Computational Neuroscience — August 2026
New York City
Yale AI Symposium — April 2026
Yale University
NeurIPS — December 2025
San Diego
Learning Meets Geometry, Graphs, and Networks — April 2025
Flatiron Institute
NYAS Annual Machine Learning Symposium — October 2024
New York City
PQG Conference — October 2024
Harvard University
ICML — July 2024
Vienna
FDS Workshop: Theory and Practice of Foundation Models — October 2023
Yale University
FRG Workshop on Stability, Moduli Spaces and Applications — December 2019
University of Illinois at Chicago
Facets of Algebraic Geometry — October 2019
University of Michigan
Algebraic Geometry Northeastern Series (AGNES) — September 2019
Boston College
Arithmetic & Algebraic Geometry in Ann Arbor — August 2019
University of Michigan
Algebraic Geometry Northeastern Series (AGNES) — March 2019
University of Massachusetts Amherst
Midwest Algebraic Geometry Graduate Conference (MAGGC) — March 2019
University of Illinois at Chicago
FRG Workshop on Moduli Spaces of Sheaves and Bridgeland Stability — December 2018
University of Illinois at Chicago
MSRI Summer Graduate Schools, Derived Categories — June 2018
MSRI
Algebraic Geometry Northeastern Series (AGNES) — September 2018
Brown University
Midwest Algebraic Geometry Graduate Conference (MAGGC) — May 2018
University of Illinois at Chicago
Algebraic Geometry Northeastern Series (AGNES) — April 2018
Rutgers University
FRG Workshop on Moduli of Sheaves and Strange Duality — December 2017
Pennsylvania State University
Algebraic Geometry Northeastern Series (AGNES) — October 2017
Northeastern University
Geometry of Moduli Spaces — May 2017

University of California, San Diego

Tokyo-Princeton Algebraic Geometry Conference — May 2016
Princeton University

Algebraic Geometry Northeastern Series (AGNES) — April 2016
Yale University

Research Visits

New York University — May 2026 - Present

Indiana University — November 2018

Service

Etale Cohomology Reading Group, Organizer — May 2018 - December 2018

Teaching

Pennsylvania State University

Math 220 Matrices, Instructor	Fall 2019
Math 111 Techniques of Calculus II, Instructor	Spring 2019
Math 497A Honors MASS Program, Polynomial Interpolation, Teaching Assistant	Fall 2018
Math 220 Matrices, Instructor	Spring 2018
Math 251 Ordinary and Partial Differential Equations, Instructor	Spring 2017
Math 251 Ordinary and Partial Differential Equations, Instructor	Fall 2016
Math 251 Ordinary and Partial Differential Equations, Instructor	Spring 2016
Math 251 Ordinary and Partial Differential Equations, Instructor	Fall 2015
Math 111 Techniques of Calculus II, Instructor	Summer 2015
Math 110 Techniques of Calculus I, Teaching Assistant	Spring 2015